

Arunkarthick Ramesh

Mechanical Design and Structural Engineer

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English — professional · Tamil — native · German — A1

Mechanical design and structural engineer with eight years in elevator equipment and industrial chain. I design EN 81-20 load-bearing components in Creo and check strength in Ansys Workbench and by hand to EN 1993, including bolts, welds, and buckling. Recent work automates those bolt checks in Ansys Mechanical by reading beam and joint probes. I have also carried out sling and heavy-lift structural analysis, and a two-month design assignment in Finland.

CORE SKILLS

CAD	Creo 9 (sheet metal, assemblies, Pro/Program), AutoCAD, SolidWorks
FEA	Ansys Workbench and Mechanical: static, modal, buckling, mesh convergence; HyperMesh
Codes	EN 81-20, EN 1993 (steel, buckling, bolt and weld checks), ISO 606, ASME B29.8, BS 4116
Release	GD&T, Windchill / PDM, MathCAD, Power Apps, VBA, strength-calculation reports

EXPERIENCE

Special Engineering Specialist · KONE Elevators
Dec 2021 — Present

Shaft Mechanical & Component Engineering · Chennai, India

- Lead design and validation of EN 81-20 load-bearing elevator components for European and Chinese markets in Creo, at about three projects a month.
- Check interference, component strength, and large assemblies in Ansys Workbench and with manual calculations.
- Two-month onsite assignment in Finland on high-rise elevator design and structural analysis.
- Mentor a six-person strength-calculation team from engineering inputs through PDM release. Use 5 Whys on design issues and report status to senior stakeholders.
- Six Sigma Green Belt initiative: engineering effort from 50 hours to 3 hours per project, and cost from about €1,000 to €120, with a further 30% cost saving. Lean work with R&D cut structural-analysis effort 56% (18 hours to 8).
- IronPython scripts in Ansys Mechanical 2025 R1 read Beam Probes and Joint Probes and check bolts for slip and for EN 1993-1-8 tension, shear, and combined utilization. Bolt size on beams comes from the radius.

Design Engineer · Renold Chain India
Apr 2019 — Dec 2021

Dindigul, India · Renold Einbeck and Renold India

- Released 20–25 manufacturing drawings a month for NPI, legacy conversion, and revisions in AutoCAD, Creo, and Windchill.
- Designed sheet-metal forming tools, stamping dies, welding fixtures, and production jigs, plus segmented and hub sprockets and large assemblies.
- Prepared CSR, BOM, and EDS, using metallurgy and heat-treatment practice. Standards: ISO 606, ASME B29.8, and BS 4116.

Business Development Engineer · Renold Chain India
Jan 2019 — Apr 2019

Dindigul, India

- Worked with OEMs, dealers, and regional sales. Customer reach increased 20%.

Graduate Engineer Trainee · Renold Chain India
Jul 2018 — Dec 2018

Dindigul, India

- Industrial training in design and manufacturing for chain products.

SELECTED WORK

Jeddah Tower · elevator sling structural analysis KONE · ~120 h

- Structural analysis of sling systems: lifting configuration, integrity, and safety.

King Salman project · 12-ton heavy lift KONE · ~30 h

- Structural analysis of a 12-ton heavy-duty lift, including heavy bracket strength.

STRUCTIVA · 2D structural drawing and analysis 2026 · Personal

- Draws members, supports, and loads; solves equilibrium; plots shear force and bending moment; basic 2D truss.

AWARDS

- Innovation Award 2026 — top 10 ideas out of 169.
- Ace Performer, KONE, 2023 and 2025.

EDUCATION

B.E. Mechanical Engineering · CGPA 7.5 / 10 2014 — 2018

Sri Ramakrishna Institute of Technology (Anna University), Coimbatore

- Member, SAE. Member, The Institute of Indian Foundrymen.

CERTIFICATIONS

- Six Sigma Yellow Belt — KONE
- HyperMesh for FEA, plastics and sheet metal — Skill-Lync, May 2023
- Geometric Dimensioning & Tolerancing: Basics — Udemy, May 2020
- Sheet Metal Design — Udemy, March 2020